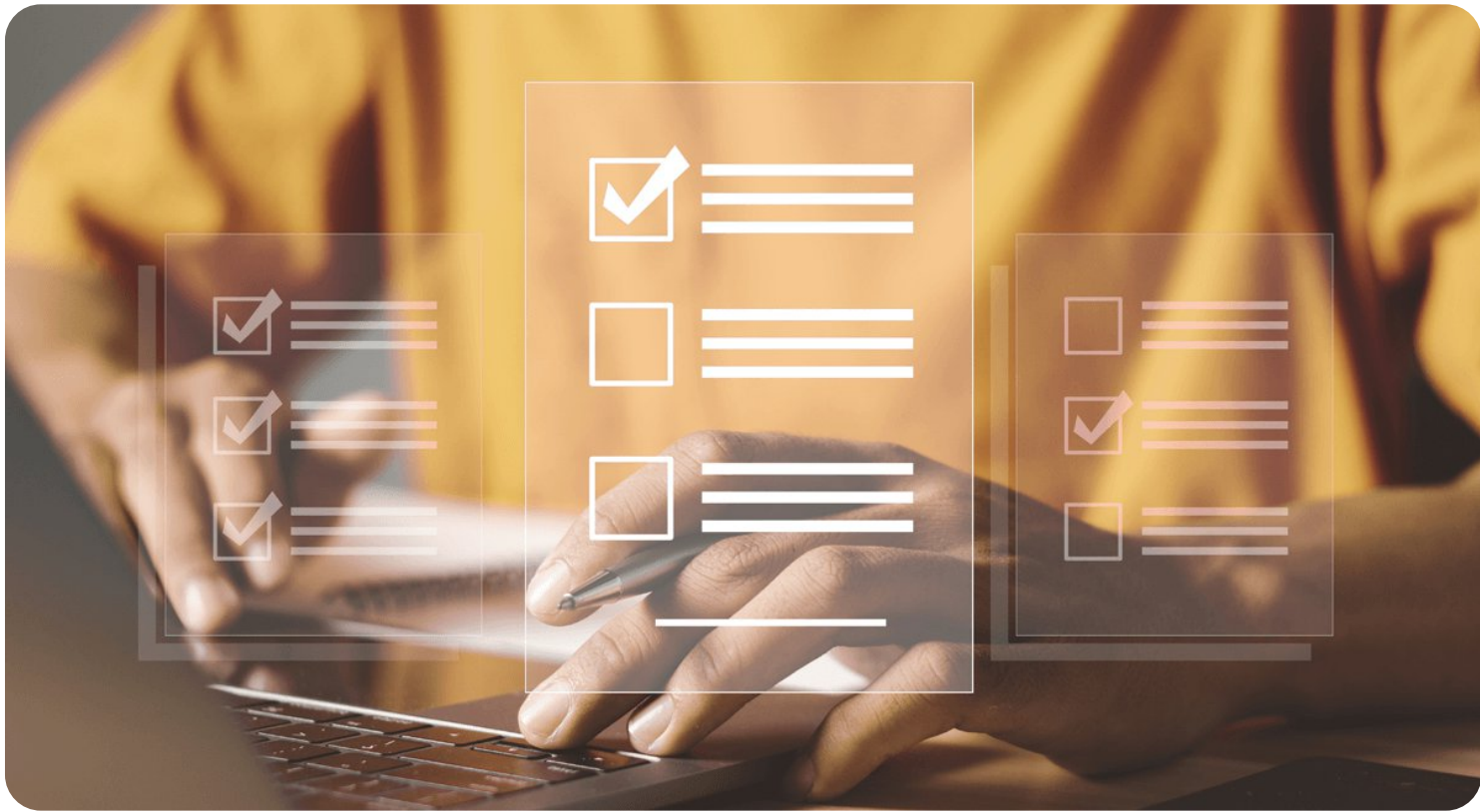




Semana 1

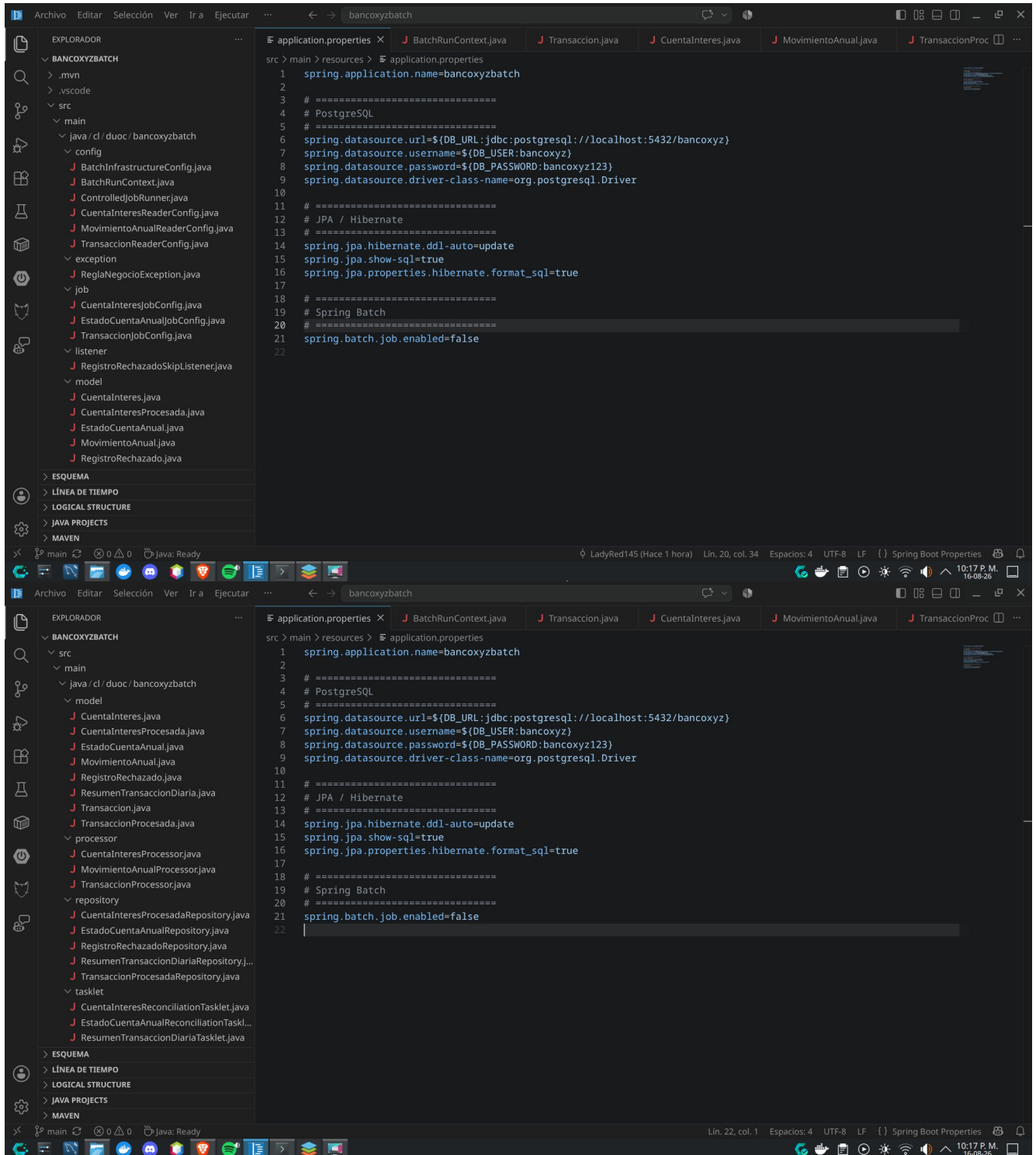
Desarrollo Backend III (PBY2203)

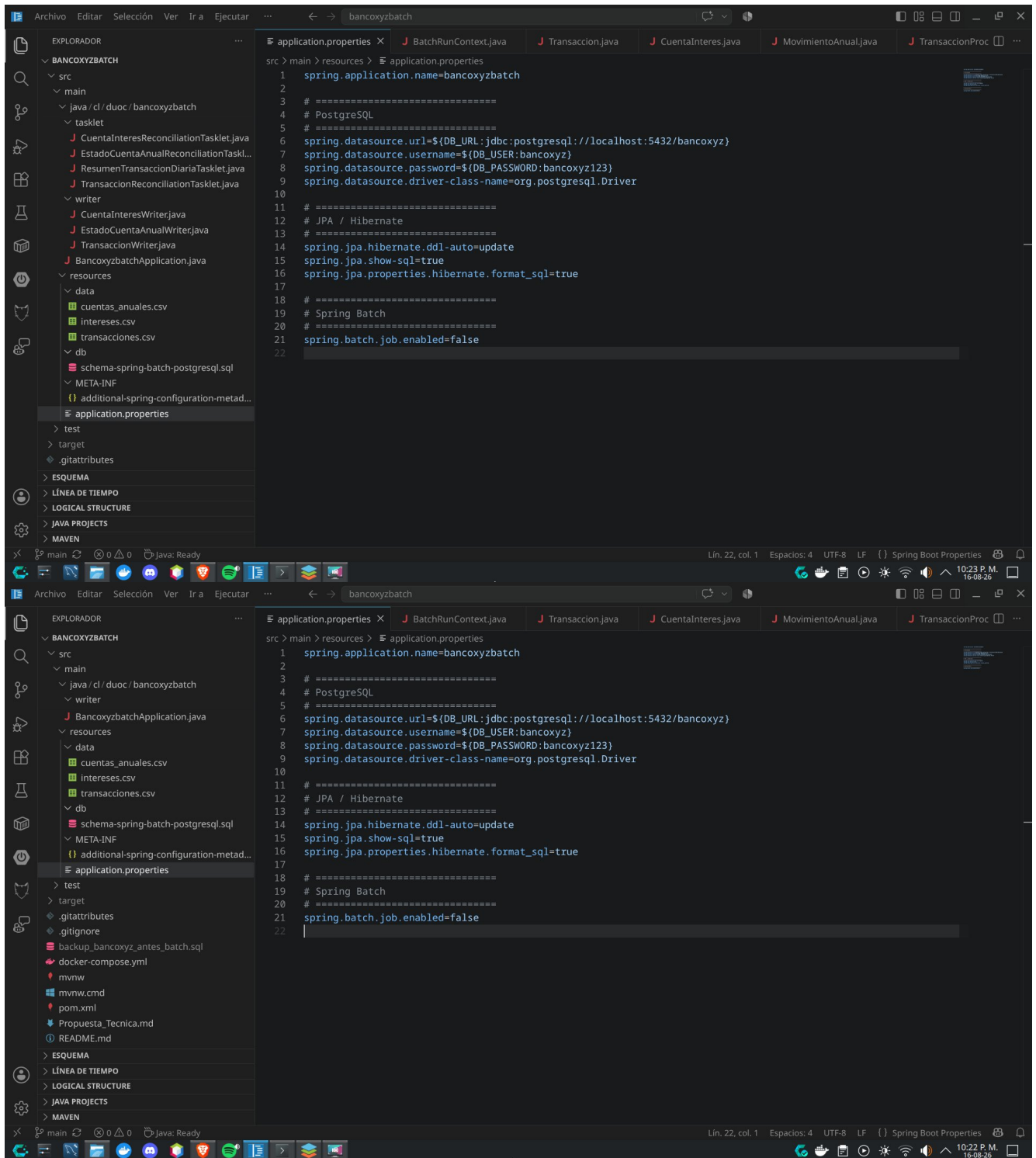
Nombre estudiantes:	• Natalia Alvarado • Egor Llanapichun
Asignatura: Desarrollo Backend III	Carrera: Analista Programador Computacional
Profesor: Alonso Castillo	Fecha: 16-08-2026

Índice del PDF de evidencias

1. Estructura del proyecto
2. Pruebas Maven y compilación
3. Ejecución de transaccionJob
4. Resultados de transacciones y resumen diario
5. Ejecución de cuentaInteresJob
6. Resultados del cálculo de intereses
7. Ejecución de estadoCuentaAnualJob
8. Resultados de estados de cuenta anuales
9. Manejo de errores y registros rechazados
10. Historial final de Jobs y Steps
11. Repositorio de GitHub

1. Estructura del proyecto






```
bancoxyzbatch : fish — Console
Nueva pestaña  Dividir vista
Copiar  Pegar  Buscar

bancoxyzbatch : docker x  bancoxyzbatch : fish x
~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main
> ./mvnw clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----[ cl.duoc:bancoxyzbatch ]-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ bancoxyzbatch ---
[INFO] Deleting /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/target
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ bancoxyzbatch ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 5 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ bancoxyzbatch ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 35 source files with javac [debug parameters release 21] to target/classes
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ bancoxyzbatch ---
[INFO] skip non existing resourceDirectory /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ bancoxyzbatch ---
[INFO] Recompiling the module because of changed dependency.
[INFO] Compiling 1 source file with javac [debug parameters release 21] to target/test-classes
[INFO]
[INFO] --- surefire:3.5.6:test (default-test) @ bancoxyzbatch ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running cl.duoc.bancoxyzbatch.BancoxyzbatchApplicationTests
22:28:21.452 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [cl.duoc.bancoxyzbatch.BancoxyzbatchApplicationTests]: BancoxyzbatchApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
22:28:21.553 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration cl.duoc.bancoxyzbatch.BancoxyzbatchApplication for test class cl.duoc.bancoxyzbatch.BancoxyzbatchApplicationTests
22:28:21.613 [main] INFO org.springframework.test.context.support.AnnotationConfigContextLoaderUtils -- Could not detect default configuration classes for test class [cl.duoc.bancoxyzbatch.BancoxyzbatchApplicationTests]: BancoxyzbatchApplicationTests does not declare any static, non-private, non-final, nested classes annotated with @Configuration.
22:28:21.615 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper -- Found @SpringBootConfiguration cl.duoc.bancoxyzbatch.BancoxyzbatchApplication for test class cl.duoc.bancoxyzbatch.BancoxyzbatchApplicationTests
22:28:21.669 [main] INFO org.springframework.boot.devtools.restart.RestartApplicationListener -- Restart disabled due to context in which it is running

[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.783 s
[INFO] Finished at: 2026-08-16T22:28:25-04:00
[INFO]
~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 9s

bancoxyzbatch : fish — Console
Nueva pestaña  Dividir vista
Copiar  Pegar  Buscar

bancoxyzbatch : docker x  bancoxyzbatch : fish x
2026-08-16T22:28:23.394-04:00 INFO 120247 --- [bancoxyzbatch] [ main] com.zaxxer.hikari.pool.HikariPool : HikariPool-1 - Added connection org.postgresql.jdbc.PgCon
nection@2f8f40cc
2026-08-16T22:28:23.395-04:00 INFO 120247 --- [bancoxyzbatch] [ main] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Start completed.
2026-08-16T22:28:23.478-04:00 INFO 120247 --- [bancoxyzbatch] [ main] org.hibernate.org.connections.pooling : HHH10001005: Database Info:
Database driver: PostgreSQL JDBC Driver
Database dialect: PostgreSQLDialect
Database version: 17.11
Default catalog/schema: bancoxyz/public
Autocommit mode: undefined/unknown
Isolation level: READ_COMMITTED [default READ_COMMITTED]
JDBC fetch size: none
Pool: DataSourceConnectionProvider
Minimum pool size: undefined/unknown
Maximum pool size: undefined/unknown
2026-08-16T22:28:24.240-04:00 INFO 120247 --- [bancoxyzbatch] [ main] org.hibernate.orm.core : HHH000489: No JTA platform available (set 'hibernate.tran
saction.jta.platform' to enable JTA platform integration)
2026-08-16T22:28:24.288-04:00 INFO 120247 --- [bancoxyzbatch] [ main] j.LocalContainerEntityManagerFactoryBean : Initialized JPA EntityManagerFactory for persistence unit
'default'
2026-08-16T22:28:24.430-04:00 INFO 120247 --- [bancoxyzbatch] [ main] o.s.d.j.r.query.QueryEnhancerFactories : Hibernate is in classpath; If applicable, HQL parser will
be used.
2026-08-16T22:28:25.247-04:00 INFO 120247 --- [bancoxyzbatch] [ main] c.d.b.BancoxyzbatchApplicationTests : Started BancoxyzbatchApplicationTests in 3.587 seconds (p
rocess running for 3.561)
[BATCH] Ningún Job solicitado.
Mockito is currently self-attaching to enable the inline-mock-maker. This will no longer work in future releases of the JDK. Please add Mockito as an agent to your build as describe
d in Mockito's documentation: https://javadoc.io/doc/org.mockito/mockito-core/latest/org.mockito/org/mockito/Mockito.html#0.3
OpenJDK 64-Bit Server VM warning: Sharing is only supported for boot loader classes because bootstrap classpath has been appended
WARNING: Java agent has been loaded dynamically (/home/natalia/.m2/repository/net/bytebuddy/byte-buddy-agent/1.18.10/byte-buddy-agent-1.18.10.jar)
WARNING: If a serviceability tool is in use, please run with -XX:+EnableDynamicAgentLoading to hide this warning
WARNING: If a serviceability tool is not in use, please run with -Djdk.instrument.traceUsage for more information
WARNING: Dynamic loading of agents will be disallowed by default in a future release
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.406 s -- in cl.duoc.bancoxyzbatch.BancoxyzbatchApplicationTests
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 8.783 s
[INFO] Finished at: 2026-08-16T22:28:25-04:00
[INFO]
~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 9s
```

3. Ejecución de transaccionJob

```
bancoxyzbatch : fish — Konsole

Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar  ☰

bancoxyzbatch : docker  bancoxyzbatch : fish

~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 9s
> ./mvnw spring-boot:run \
  -Dspring-boot.run.arguments="--app.batch.job=transaccionJob --app.batch.run-id=18"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< cl.duoc:bancoxyzbatch >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO]
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ bancoxyzbatch >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ bancoxyzbatch ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 5 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ bancoxyzbatch ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ bancoxyzbatch ---
[INFO] skip non existing resourceDirectory /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ bancoxyzbatch ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ bancoxyzbatch <<<
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ bancoxyzbatch ---
[INFO] Attaching agents: []

  ____ _
 / ___ \| | | |
/ /   \| |_| |
\ \   /| | | |
 \ \_/ \| |_| |
  \___|_|_|_|_|

:: Spring Boot ::
              (v4.1.0)

2026-08-16T22:31:21.136-04:00 INFO 122084 --- [bancoxyzbatch] [ restartedMain] c.d.b.BancoxyzbatchApplication : Starting BancoxyzbatchApplication using Java 21.0.12 with
PID 122084 (/home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/target/classes started by natalia in /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxy
zbatch)
2026-08-16T22:31:21.132-04:00 INFO 122084 --- [bancoxyzbatch] [ restartedMain] c.d.b.BancoxyzbatchApplication : No active profile set, falling back to 1 default profile:
"default"

insert
into
  resumen_transacciones_diarias
(cantidad_transacciones, saldo_neto, total_creditos, total_debitos, ultima_instancia_id, fecha)
values
  (?, ?, ?, ?, ?, ?)
Hibernate:
insert
into
  resumen_transacciones_diarias
(cantidad_transacciones, saldo_neto, total_creditos, total_debitos, ultima_instancia_id, fecha)
values
  (?, ?, ?, ?, ?, ?)
Hibernate:
insert
into
  resumen_transacciones_diarias
(cantidad_transacciones, saldo_neto, total_creditos, total_debitos, ultima_instancia_id, fecha)
values
  (?, ?, ?, ?, ?, ?)
Hibernate:
insert
into
  resumen_transacciones_diarias
(cantidad_transacciones, saldo_neto, total_creditos, total_debitos, ultima_instancia_id, fecha)
values
  (?, ?, ?, ?, ?, ?)
2026-08-16T22:31:24.338-04:00 INFO 122084 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Step: [resumenTransaccionDiariaStep] executed in 95ms
2026-08-16T22:31:24.368-04:00 INFO 122084 --- [bancoxyzbatch] [ restartedMain] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=transaccionJob]] completed with th
e following parameters: [{JobParameter[name='run.id', value=18, type=class java.lang.Long, identifying=true]] and the following status: [COMPLETED] in 418ms
[BATCH] JobExecution ID: 18
[BATCH] Estado final: COMPLETED
[BATCH] Exit status: exitCode=COMPLETED;exitDescription=;exitException=null
2026-08-16T22:31:24.374-04:00 INFO 122084 --- [bancoxyzbatch] [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'de
fault'
2026-08-16T22:31:24.376-04:00 INFO 122084 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-08-16T22:31:24.379-04:00 INFO 122084 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] -----
[INFO] Total time: 4.886 s
[INFO] Finished at: 2026-08-16T22:31:24-04:00
[INFO]
[INFO] -----

~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main
```

4. Resultados de transacciones y resumen diario

bancoxyzbatch : docker — Konsole

Nueva pestaña

Dividir vista

Copiar

Pegar

Buscar

~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main
\$ docker exec -it bancoxyz-postgres \psql -U bancoxyz -d bancoxyz
psql (17.11 (Debian 17.11-1.pgdg13+2))
Type "help" for help.

bancoxyz=# \set pager off
Pager usage is off.
bancoxyz=# SELECT
id,
fecha,
monto,
tipo,
estado,
observacion,
activo
FROM transacciones_procesadas
ORDER BY id;

id	fecha	monto	tipo	estado	observacion	activo
1	2024-01-01	1000.00	debito	PROCESADO	Transacción válida	t
2	2024-01-02	1500.00	credito	PROCESADO	Transacción válida	t
3	2024-01-03	200.00	debito	CORREGIDO	Monto negativo de débito normalizado a valor absoluto	t
4	2024-01-03	0.00	debito	RECHAZADO	Transacción con monto igual a cero	t
5	2024-01-04	800.00	credito	PROCESADO	Transacción válida	t
6	2024-01-05	700.00	debito	PROCESADO	Transacción válida	t
7	2024-01-06	1200.00	credito	PROCESADO	Transacción válida	t
8	2024-01-05	700.00	debito	DUPLICADO	Posible duplicado de contenido. Coincide con la transacción ID 6	t
9	2024-01-07	3000.00	debito	PROCESADO	Transacción válida	t
10	2024-01-08	1000.00	credito	PROCESADO	Transacción válida	t

(10 rows)

bancoxyz=# SELECT
fecha,
cantidad_transacciones,
total_credits,
total_debitos,
saldo_neto
FROM resumen_transacciones_diarias
ORDER BY fecha;

fecha	cantidad_transacciones	total_credits	total_debitos	saldo_neto
2024-01-01	1	0.00	1000.00	-1000.00
2024-01-02	1	1500.00	0.00	1500.00
2024-01-03	1	0.00	200.00	-200.00
2024-01-04	1	800.00	0.00	800.00

bancoxyzbatch : docker — Konsole

Nueva pestaña

Dividir vista

Copiar

Pegar

Buscar

bancoxyzbatch : docker — Konsole
Pager usage is off.
bancoxyz=# SELECT
id,
fecha,
monto,
tipo,
estado,
observacion,
activo
FROM transacciones_procesadas
ORDER BY id;

id	fecha	monto	tipo	estado	observacion	activo
1	2024-01-01	1000.00	debito	PROCESADO	Transacción válida	t
2	2024-01-02	1500.00	credito	PROCESADO	Transacción válida	t
3	2024-01-03	200.00	debito	CORREGIDO	Monto negativo de débito normalizado a valor absoluto	t
4	2024-01-03	0.00	debito	RECHAZADO	Transacción con monto igual a cero	t
5	2024-01-04	800.00	credito	PROCESADO	Transacción válida	t
6	2024-01-05	700.00	debito	PROCESADO	Transacción válida	t
7	2024-01-06	1200.00	credito	PROCESADO	Transacción válida	t
8	2024-01-05	700.00	debito	DUPLICADO	Posible duplicado de contenido. Coincide con la transacción ID 6	t
9	2024-01-07	3000.00	debito	PROCESADO	Transacción válida	t
10	2024-01-08	1000.00	credito	PROCESADO	Transacción válida	t

(10 rows)

bancoxyz=# SELECT
fecha,
cantidad_transacciones,
total_credits,
total_debitos,
saldo_neto
FROM resumen_transacciones_diarias
ORDER BY fecha;

fecha	cantidad_transacciones	total_credits	total_debitos	saldo_neto
2024-01-01	1	0.00	1000.00	-1000.00
2024-01-02	1	1500.00	0.00	1500.00
2024-01-03	1	0.00	200.00	-200.00
2024-01-04	1	800.00	0.00	800.00
2024-01-05	1	0.00	700.00	-700.00
2024-01-06	1	1200.00	0.00	1200.00
2024-01-07	1	0.00	3000.00	-3000.00
2024-01-08	1	1000.00	0.00	1000.00

(8 rows)

bancoxyz=#

5. Ejecución de cuentaInteresJob

```
bancoxyzbatch: fish — Konsole

Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar  ☰

bancoxyzbatch: docker  x  bancoxyzbatch: fish  x
~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 4m
> ./mvnw spring-boot:run \
  -Dspring-boot.run.arguments="--app.batch.job=cuentaInteresJob --app.batch.run-id=19"
[INFO] Scanning for projects...
[INFO] -----< cl.duoc:bancoxyzbatch >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ bancoxyzbatch >>>
[INFO] --- resources:3.5.0:resources (default-resources) @ bancoxyzbatch ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 5 resources from src/main/resources to target/classes
[INFO] --- compiler:3.15.0:compile (default-compile) @ bancoxyzbatch ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.5.0:testResources (default-testResources) @ bancoxyzbatch ---
[INFO] skip non existing resourceDirectory /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/src/test/resources
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ bancoxyzbatch ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ bancoxyzbatch <<<
[INFO] --- spring-boot:4.1.0:run (default-cli) @ bancoxyzbatch ---
[INFO] Attaching agents: []

:: Spring Boot ::
(v4.1.0)

2026-08-16T22:38:10.054-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] c.d.b.BancoxyzbatchApplication : Starting BancoxyzbatchApplication using Java 21.0.12 with
PID 126372 (/home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/target/classes started by natalia in /home/natalia/Archivos/Desarrollo Backend III/bancoxyz
batch)
2026-08-16T22:38:10.056-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] c.d.b.BancoxyzbatchApplication : No active profile set, falling back to 1 default profile:
"default"
2026-08-16T22:38:10.105-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.a

cuentas_intereses
set
  activo=?,
  estado=?,
  interes_calculado=?,
  nombres=?,
  observacion=?,
  saldo_final=?,
  saldo_inicial=?,
  tasa_interes=?,
  tipo=?,
  ultima_instancia_id=?
where
  cuenta_id=?
2026-08-16T22:38:13.302-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Step: [cuentaInteresStep] executed in 130ms
2026-08-16T22:38:13.348-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Executing step: [cuentaInteresReconciliationStep]
Hibernate:
update
  cuentas_intereses cipi_0
set
  activo=false
where
  cipi_0.activo=true
and (
  cipi_0.ultima_instancia_id is null
  or cipi_0.ultima_instancia_id<=?
)
2026-08-16T22:38:13.407-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Step: [cuentaInteresReconciliationStep] executed in 65ms
2026-08-16T22:38:13.448-04:00 INFO 126372 --- [bancoxyzbatch] [ restartedMain] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=cuentaInteresJob]] completed with
the following parameters: [{JobParameter{name='run.id', value=19, type=class java.lang.Long, identifying=true}}] and the following status: [COMPLETED] in 288ms
[BATCH] JobExecution ID: 19
[BATCH] Estado final: COMPLETED
[BATCH] Exit status: exitCode=COMPLETED;exitDescription=;exitException=null
2026-08-16T22:38:13.456-04:00 INFO 126372 --- [bancoxyzbatch] [onShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'de
fault'
2026-08-16T22:38:13.460-04:00 INFO 126372 --- [bancoxyzbatch] [onShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-08-16T22:38:13.463-04:00 INFO 126372 --- [bancoxyzbatch] [onShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.458 s
[INFO] Finished at: 2026-08-16T22:38:13-04:00
[INFO] -----

~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 6s
>
```

6. Resultados de intereses

```
bancoxyzbatch : docker -- Konsole
Nueva pestaña  Dividir vista
bancoxyzbatch : docker x bancoxyzbatch : docker x
[BATCH] JobExecution ID: 19
[BATCH] Estado final: COMPLETED
[BATCH] Exit status: exitCode=COMPLETED;exitDescription=;exitException=null
2026-08-16T22:38:13.456-04:00 INFO 126372 --- [bancoxyzbatch] [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'de
fault'
2026-08-16T22:38:13.460-04:00 INFO 126372 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-08-16T22:38:13.463-04:00 INFO 126372 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.458 s
[INFO] Finished at: 2026-08-16T22:38:13-04:00
[INFO] -----

~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 6s
> docker exec -it bancoxyz-postgres \
  psql -U bancoxyz -d bancoxyz
psql (17.11 (Debian 17.11-1.pgdg13+2))
Type "help" for help.

bancoxyz=# \pset pager off
Pager usage is off.
bancoxyz=# SELECT
  cuenta_id,
  saldo_inicial,
  tipo,
  tasa_interes,
  interes_calculado,
  saldo_final,
  estado,
  observacion
FROM cuentas_intereses
ORDER BY cuenta_id;
cuenta_id | saldo_inicial | tipo | tasa_interes | interes_calculado | saldo_final | estado | observacion
-----
101 | 5000.00 | ahorro | 0.0100 | 50.00 | 5050.00 | PROCESADO | Interés mensual calculado con tasa de ahorro del 1%
102 | 8000.00 | prestamo | 0.0200 | 160.00 | 8160.00 | PROCESADO | Interés mensual calculado con tasa de préstamo del 2%
103 | 12000.00 | prestamo | 0.0200 | 240.00 | 12240.00 | PROCESADO | Interés mensual calculado con tasa de préstamo del 2%
104 | 0.00 | ahorro | 0.0000 | 0.00 | 0.00 | SIN INTERES | Saldo igual a cero: no genera interés en el período
105 | 7000.00 | hipoteca | 0.0000 | 0.00 | 7000.00 | RECHAZADO | Tipo de cuenta no soportado para cálculo de intereses: hipoteca
106 | 5000.00 | ahorro | 0.0100 | 50.00 | 5050.00 | PROCESADO | Interés mensual calculado con tasa de ahorro del 1%
107 | 15000.00 | prestamo | 0.0200 | 300.00 | 15300.00 | PROCESADO | Interés mensual calculado con tasa de préstamo del 2%
108 | 10000.00 | ahorro | 0.0100 | 100.00 | 10100.00 | PROCESADO | Interés mensual calculado con tasa de ahorro del 1%
(8 rows)

bancoxyz=#
```

7. Ejecución de estadoCuentaAnualJob

```
bancoxyzbatch : fish — Konsole

Nueva pestaña  Dividir vista  Copiar  Pegar  Buscar  ☰

bancoxyzbatch : docker  bancoxyzbatch : fish
~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 3m 18s
> ./mvnw spring-boot:run \
  -Dspring-boot.run.arguments="--app.batch.job=estadoCuentaAnualJob --app.batch.run-id=20"
[INFO] Scanning for projects...
[INFO]
[INFO] -----< cl.duoc:bancoxyzbatch >-----
[INFO] Building 0.0.1-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] >>> spring-boot:4.1.0:run (default-cli) > test-compile @ bancoxyzbatch >>>
[INFO]
[INFO] --- resources:3.5.0:resources (default-resources) @ bancoxyzbatch ---
[INFO] Copying 1 resource from src/main/resources to target/classes
[INFO] Copying 5 resources from src/main/resources to target/classes
[INFO]
[INFO] --- compiler:3.15.0:compile (default-compile) @ bancoxyzbatch ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] --- resources:3.5.0:testResources (default-testResources) @ bancoxyzbatch ---
[INFO] skip non existing resourceDirectory /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/src/test/resources
[INFO]
[INFO] --- compiler:3.15.0:testCompile (default-testCompile) @ bancoxyzbatch ---
[INFO] Nothing to compile - all classes are up to date.
[INFO]
[INFO] <<< spring-boot:4.1.0:run (default-cli) < test-compile @ bancoxyzbatch <<<
[INFO]
[INFO] --- spring-boot:4.1.0:run (default-cli) @ bancoxyzbatch ---
[INFO] Attaching agents: []

:: Spring Boot ::
(v4.1.0)

2026-08-16T22:49:07.748-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] c.d.b.BancoxyzbatchApplication : Starting BancoxyzbatchApplication using Java 21.0.12 with
PID 132571 (/home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch/target/classes started by natalia in /home/natalia/Archivos/Desarrollo Backend III/Semana 1/bancoxy
zbatch)
2026-08-16T22:49:07.752-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] c.d.b.BancoxyzbatchApplication : No active profile set, falling back to 1 default profile:
"default"
2026-08-16T22:49:07.822-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] .e.DevToolsPropertyDefaultsPostProcessor : Devtools property defaults active! Set 'spring.devtools.a

update
estados_cuenta_anuales
set
activo=?,
anio=?,
cuenta_id=?,
saldo_anual=?,
total_compras=?,
total_depositos=?,
total_retiros=?,
ultima_instancia_id=?
where
id=?
2026-08-16T22:49:10.983-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Step: [estadoCuentaAnualStep] executed in 248ms
2026-08-16T22:49:10.950-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Executing step: [estadoCuentaAnualReconciliationStep]
Hibernate:
update
estados_cuenta_anuales eca1_0
set
activo=false
where
eca1_0.activo=true
and (
eca1_0.ultima_instancia_id is null
or eca1_0.ultima_instancia_id<=?
)
2026-08-16T22:49:10.977-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] o.s.batch.core.step.AbstractStep : Step: [estadoCuentaAnualReconciliationStep] executed in 3
3ms
2026-08-16T22:49:11.021-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=estadoCuentaAnualJob]] completed w
ith the following parameters: [{@JobParameter(name='run.id', value=20, type=class java.lang.Long, identifying=true)}] and the following status: [COMPLETED] in 382ms
[BATCH] JobExecution ID: 20
[BATCH] Estado final: COMPLETED
[BATCH] Exit status: exitCode=COMPLETED;exitDescription=;exitException=null
2026-08-16T22:49:11.052-04:00 INFO 132571 --- [bancoxyzbatch] [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'de
fault'
2026-08-16T22:49:11.065-04:00 INFO 132571 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-08-16T22:49:11.070-04:00 INFO 132571 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 5.504 s
[INFO] Finished at: 2026-08-16T22:49:11-04:00
[INFO]
~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 6s
```


8. Resultados del estado anual

```
bancoxyzbatch : docker -- Konsole
Nueva pestaña Dividir vista Copiar Pegar Buscar
bancoxyzbatch : docker x bancoxyzbatch : docker x
2026-08-16T22:49:11.021-04:00 INFO 132571 --- [bancoxyzbatch] [ restartedMain] o.s.b.c.l.s.TaskExecutorJobLauncher : Job: [SimpleJob: [name=estadoCuentaAnualJob]] completed w
ith the following parameters: [{JobParameter[name='run.id', value=20, type=class java.lang.Long, identifying=true]]} and the following status: [COMPLETED] in 382ms
[BATCH] JobExecution ID: 20
[BATCH] Estado final: COMPLETED
[BATCH] Exit status: exitCode=COMPLETED;exitDescription=;exitException=null
2026-08-16T22:49:11.052-04:00 INFO 132571 --- [bancoxyzbatch] [ionShutdownHook] j.LocalContainerEntityManagerFactoryBean : Closing JPA EntityManagerFactory for persistence unit 'de
fault'
2026-08-16T22:49:11.065-04:00 INFO 132571 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown initiated...
2026-08-16T22:49:11.070-04:00 INFO 132571 --- [bancoxyzbatch] [ionShutdownHook] com.zaxxer.hikari.HikariDataSource : HikariPool-1 - Shutdown completed.
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 5.504 s
[INFO] Finished at: 2026-08-16T22:49:11-04:00
[INFO] -----

~/Archivos/Desarrollo Backend III/Semana 1/bancoxyzbatch main 6s
> docker exec -it bancoxyz-postgres \
    psql -U bancoxyz -d bancoxyz
psql (17.11 (Debian 17.11-1.pgdg13+2))
Type "help" for help.

bancoxyz=# \pset pager off
Pager usage is off.
bancoxyz=# SELECT
    cuenta_id,
    anio,
    total_depositos,
    total_retiros,
    total_compras,
    saldo_anual,
    activo
FROM estados_cuenta_anuales
ORDER BY cuenta_id, anio;
cuenta_id | anio | total_depositos | total_retiros | total_compras | saldo_anual | activo
-----
101 | 2024 | 1000.00 | 500.00 | 0.00 | 500.00 | t
102 | 2024 | 1500.00 | 0.00 | 0.00 | 1500.00 | t
103 | 2024 | 2000.00 | 0.00 | 0.00 | 2000.00 | t
104 | 2024 | 0.00 | 0.00 | 100.00 | -100.00 | t
105 | 2024 | 2500.00 | 0.00 | 0.00 | 2500.00 | t
106 | 2024 | 3000.00 | 0.00 | 0.00 | 3000.00 | t
108 | 2024 | 2000.00 | 0.00 | 0.00 | 2000.00 | t
(7 rows)

bancoxyz=#
```

9. Manejo de errores

```
bancoxyz=# SELECT
id,
job_name,
step_name,
fase,
tipo_error,
job_instance_id
FROM registros_rechazados
ORDER BY id;
```

id	job_name	step_name	fase	tipo_error	job_instance_id
1	transaccionJob	transaccionStep	READ	FlatFileParseException	9
2	estadoCuentaAnualJob	estadoCuentaAnualStep	PROCESS	ReglaNegocioException	11
3	estadoCuentaAnualJob	estadoCuentaAnualStep	PROCESS	ReglaNegocioException	17
4	estadoCuentaAnualJob	estadoCuentaAnualStep	PROCESS	ReglaNegocioException	20

(4 rows)

```
bancoxyz=#
```

(sin mensaje)

```
bancoxyz=# SELECT
id,
job_name,
step_name,
fase,
tipo_error,
LEFT(mensaje_error, 60) AS mensaje_resumido,
job_instance_id
FROM registros_rechazados
ORDER BY id;
```

id	job_name	step_name	fase	tipo_error	mensaje_resumido	job_instance_id
1	transaccionJob	transaccionStep	READ	FlatFileParseException	Parsing error at line: 11 in resource=[class path resource [	9
2	estadoCuentaAnualJob	estadoCuentaAnualStep	PROCESS	ReglaNegocioException	El movimiento anual posee monto igual a cero	11
3	estadoCuentaAnualJob	estadoCuentaAnualStep	PROCESS	ReglaNegocioException	El movimiento anual posee monto igual a cero	17
4	estadoCuentaAnualJob	estadoCuentaAnualStep	PROCESS	ReglaNegocioException	El movimiento anual posee monto igual a cero	20

(4 rows)

```
bancoxyz=#
```

(con mensaje)

Se evidencia el registro persistente de errores técnicos de lectura y errores de negocio durante el procesamiento batch.

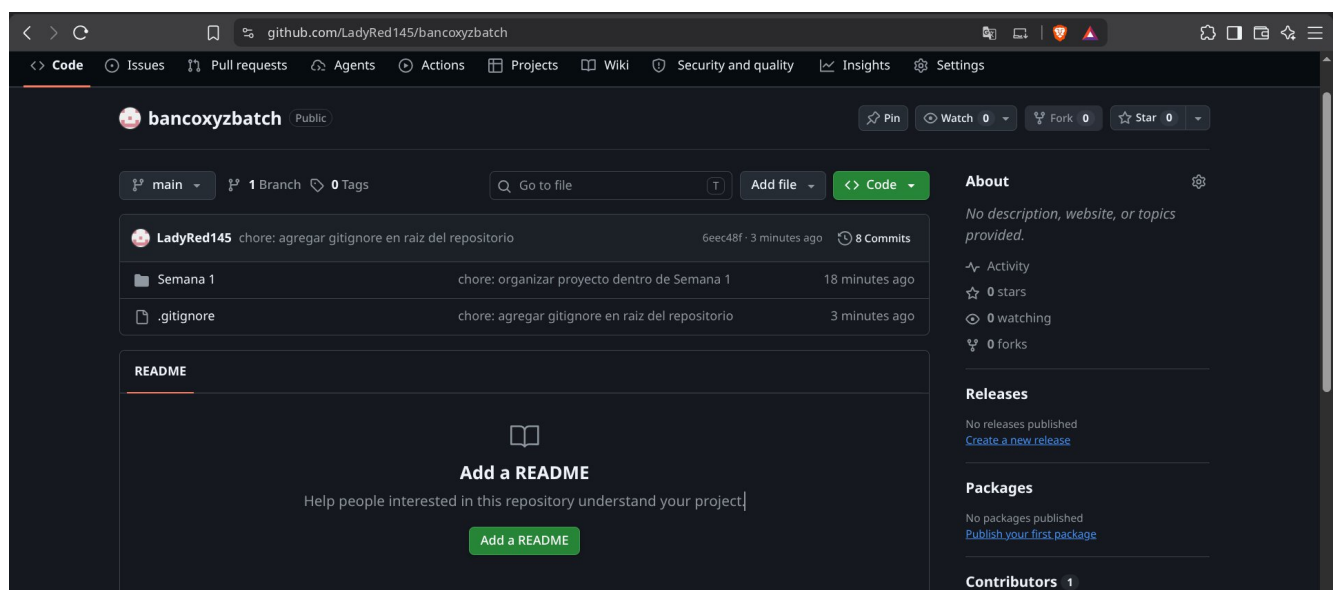
10. Historial final de Jobs y Steps

```
bancoxyzbatch : docker — Konsole
Nueva pestaña  Dividir vista
bancoxyzbatch : docker x bancoxyzbatch : docker x
bancoxyz=# SELECT
  ji.job_instance_id,
  ji.job_name,
  je.job_execution_id,
  je.status,
  je.exit_code
FROM batch_job_instance ji
JOIN batch_job_execution je
  ON ji.job_instance_id = je.job_instance_id
ORDER BY je.job_execution_id DESC
LIMIT 10;
job_instance_id | job_name | job_execution_id | status | exit_code
-----
20 | estadoCuentaAnualJob | 20 | COMPLETED | COMPLETED
19 | cuentaInteresJob | 19 | COMPLETED | COMPLETED
18 | transaccionJob | 18 | COMPLETED | COMPLETED
17 | estadoCuentaAnualJob | 17 | COMPLETED | COMPLETED
16 | cuentaInteresJob | 16 | COMPLETED | COMPLETED
15 | transaccionJob | 15 | COMPLETED | COMPLETED
14 | cuentaInteresJob | 14 | COMPLETED | COMPLETED
13 | transaccionJob | 13 | COMPLETED | COMPLETED
12 | transaccionJob | 12 | COMPLETED | COMPLETED
11 | estadoCuentaAnualJob | 11 | COMPLETED | COMPLETED
(10 rows)
```

```
bancoxyz=# SELECT
  step_execution_id,
  step_name,
  status,
  read_count,
  write_count,
  filter_count,
  read_skip_count,
  process_skip_count,
  write_skip_count
FROM batch_step_execution
ORDER BY step_execution_id DESC
LIMIT 12;
step_execution_id | step_name | status | read_count | write_count | filter_count | read_skip_count | process_skip_count | write_skip_count
-----
43 | estadoCuentaAnualReconciliationStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
42 | estadoCuentaAnualStep | COMPLETED | 9 | 8 | 0 | 0 | 1 | 0
41 | cuentaInteresReconciliationStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
40 | cuentaInteresStep | COMPLETED | 8 | 8 | 0 | 0 | 0 | 0
39 | resumenTransaccionDiariaStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
38 | transaccionReconciliationStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
37 | transaccionStep | COMPLETED | 10 | 10 | 0 | 0 | 0 | 0
36 | estadoCuentaAnualReconciliationStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
35 | estadoCuentaAnualStep | COMPLETED | 9 | 8 | 0 | 0 | 1 | 0
34 | cuentaInteresReconciliationStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
33 | cuentaInteresStep | COMPLETED | 8 | 8 | 0 | 0 | 0 | 0
32 | resumenTransaccionDiariaStep | COMPLETED | 0 | 0 | 0 | 0 | 0 | 0
(12 rows)
```

11. Repositorio de GitHub

<https://github.com/LadyRed145/bancoxyzbatch.git>





Reservados todos los derechos Fundación Instituto Profesional Duoc UC. No se permite copiar, reproducir, reeditar, descargar, publicar, emitir, difundir, de forma total o parcial la presente obra, ni su incorporación a un sistema informático, ni su transmisión en cualquier forma o por cualquier medio (electrónico, mecánico,

fotocopia, grabación u otros) sin autorización previa y por escrito de Fundación Instituto Profesional Duoc UC
La infracción de dichos derechos puede constituir un delito contra la propiedad intelectual.